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SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → ______ → Transport → Network → ______ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	20 Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	20 Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes 20 correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	15 The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-15 structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

90/100

WEB BROWSING USING OSI LAYERS – CONCEPT MAP

User Opens a Web Browser

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Reg No : 192424201

Objective

To understand how web browsing works using the OSI model and the impact when a layer fails.

Layer 7 – Application Layer
HTTP / HTTPS, Web Browser
Browser sends requests for web pages
https://

Layer 6 – Presentation Layer
SSL/TLS Encryption, Data Formatting
Encrypts and formats data
SSL/TLS

Layer 5 – Session Layer
Establishes & Maintains Session
Manages active sessions

Layer 4 – Transport Layer
TCP, Port 80/443, Reliable Delivery
Ensures complete and correct delivery
TCP 80/443

Layer 3 – Network Layer
IP Addressing & Routing
Routes packets to destination
IP

Layer 2 – Data Link Layer
MAC Address, Ethernet/Wi-Fi Frames
Transfers frames between devices

Layer 1 – Physical Layer
Cables, Wi-Fi Signals, Fiber Optics
Transmits raw bits over the medium

Internet ⟷ Web Server

ANALYSIS: EFFECT OF FAILURE IN EACH LAYER

OSI Layer	Role in Web Browsing	Effect if the Layer Fails
7 Application	Browser sends HTTP/HTTPS requests	Web pages cannot be requested or displayed.
6 Presentation	Encrypts (SSL/TLS) and formats data	HTTPS websites fail due to certificate or encryption errors.
5 Session	Creates and maintains communication sessions	User sessions disconnect or login sessions are interrupted.
4 Transport	TCP ensures reliable packet delivery	Slow loading, incomplete pages, or connection timeouts occur.
3 Network	Routes packets using IP addresses	Browser cannot reach the destination server ("Network unreachaable").
2 Data Link 1 Physical	Transfers frames using MAC addresses Transmits bits through cables or wireless signals	Devices cannot communicate over the local network or conpleivity due to internee or hardware failure.

EXAMPLE: WEB PAGE LOAD PROCESS

1. User enters URL (e.g., www.example.com)
2. Browser sends HTTP/HTTPS request to the server
3. Request is encrypted (SSL/TLS) and formated
4. Session is established between client and server
5. TCP ensures reliable data transfer using ports
6. IP routes packets across networks to server
7. Frames are sent over the local network
8. Signals are transmitted through cables/Wi-Fi
9. Web page is received and displayed in browser

KEY TAKEAWAYS

- The OSI model works like a chain from Layer 7 (Application) down to Layer 1 (Physical).
- Each layer depends on the services of the layer below it.
- If any one layer fails, communication is affected.
- Web browsing requires all layers to work together for successful data transmission.
- Troubleshooting is easier when issues are identified at the specific layer.

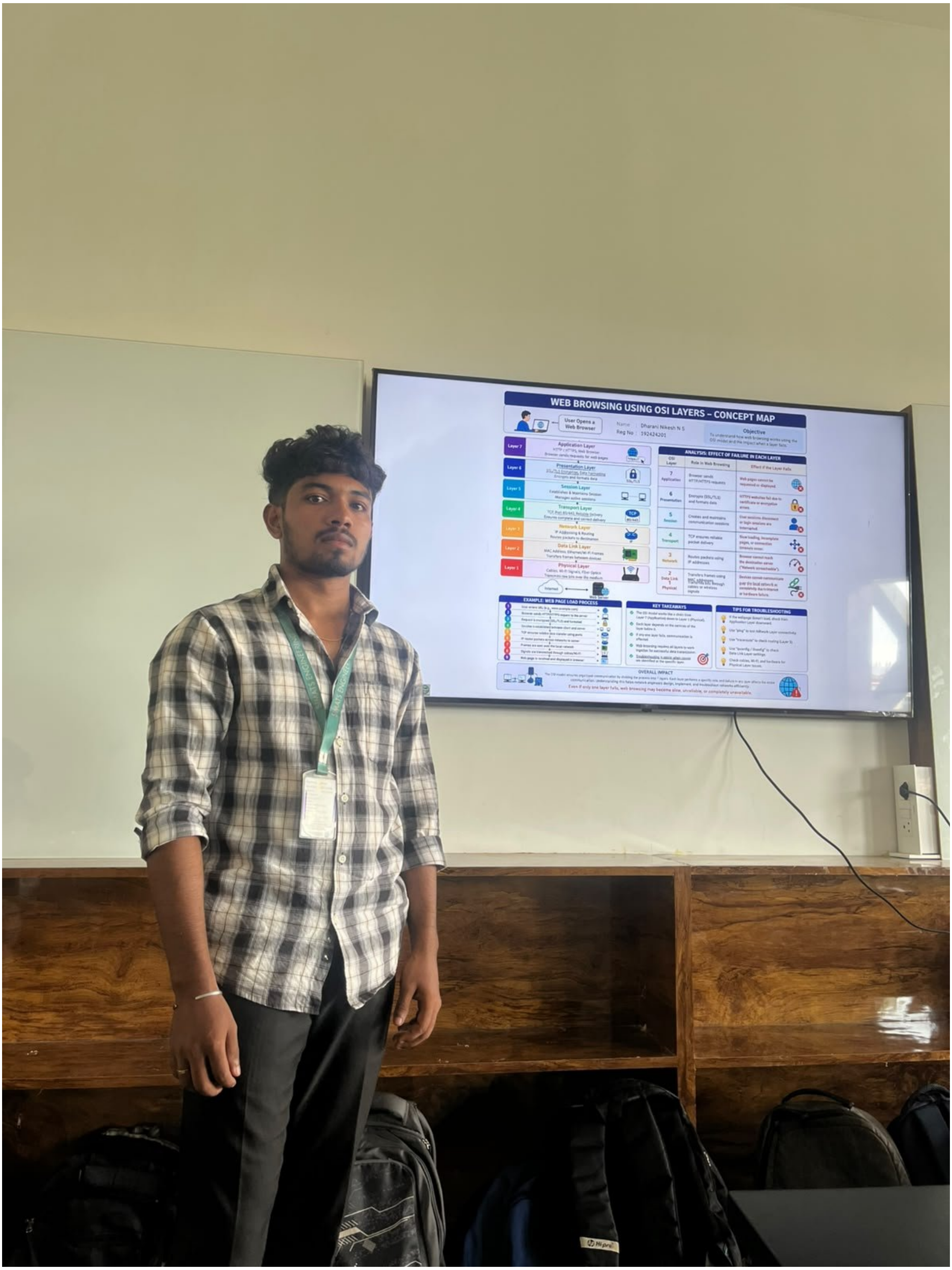
TIPS FOR TROUBLESHOOTING

- If the webpage deesn't load, check from Application Layer downward.
- Use "ping" to test Network Layer connectivity.
- Use "traceroute" to check routing (Layer 3).
- Use "ipconfig / ifconfig" to check Data Link Layer settings.
- Check cables, Wi-Fi, and hardware for Physical Layer issues.

OVERALL IMPACT

The OSI model ensures organized communication by dividing the process into 7 layers. Each layer performs a specific role, and failure in any layer affects the entire communication. Understanding this helps network engineers design, implement, and troubleshoot networks efficiently.

Even if only one layer fails, web browsing may become slow, unreliable, or completely unavailable.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